Self-Healing EC2 Fleet — 100 Real-World Use Cases

A capstone architecture: ALB health checks → ASG lifecycle hooks → terminate/launch Lambda handlers → SSM readiness verification → SNS notification.

Each entry maps a concrete failure mode to how the control loop resolves it automatically, without human intervention.

E-Commerce & Retail

1. **Black Friday traffic spike.** One instance chokes under load and starts failing health checks while the rest of the fleet holds; it's cycled out instead of dragging down the whole tier.
2. **Checkout service memory leak.** A Node/Java checkout process degrades over days; /health times out, instance is drained (in-flight carts preserved) and replaced.
3. **Bad third-party payment SDK update.** One instance's outbound calls to a payment gateway start hanging, tying up threads until health checks fail; auto-replaced before it backs up the whole queue.
4. **Product search index corruption on one node.** A single instance serves stale/broken search results; if wired to an app-level health check, it's pulled from rotation.
5. **Flash-sale bot traffic overwhelming a subset of instances.** Targeted load skews unevenly across the ASG; unhealthy nodes are replaced while WAF (once added) blocks the bots.
6. **Cart session store connection exhaustion.** An instance runs out of DB connections and stops responding; drained and replaced, session state archived to S3 first.
7. **Regional inventory sync failure.** An instance pinned to a broken read replica serves wrong stock counts; caught by a custom health check and cycled.
8. **Post-deploy canary instance going bad.** A rolling deploy pushes a crashing build to a subset; those instances fail health checks and get replaced (with alerting so the deploy itself still gets caught).
9. **Holiday-season kernel/OS-level host degradation.** Underlying AWS hardware issue causes EC2 status check failures independent of app health; ASG replaces the instance automatically.
10. **Multi-AZ failover during a zone-level event.** One AZ degrades; min-3-instance multi-AZ config keeps 99.95% availability while unhealthy nodes in the affected AZ are replaced.

SaaS & B2B Platforms

1. **Tenant-specific workload causing noisy-neighbor CPU starvation.** One instance pegged at 100% CPU by a single heavy tenant fails health checks and is replaced, isolating the blast radius.
2. **Background job worker deadlock.** A queue-consuming instance hangs on a poison message; health check fails, instance recycled, message requeued.
3. **License/auth token expiry bug on one node.** A stale cached credential causes 500s from one instance only; auto-replaced, fresh instance pulls a valid token via bootstrap.
4. **API gateway backend instance running out of file descriptors.** Long-running process leaks FDs over weeks; self-healing resets the clock without a scheduled maintenance window.
5. **Customer-reported "random 502s".** Intermittent failures traced to one flapping instance; ALB health checks catch what ad-hoc monitoring missed, and it's replaced before the next support ticket.
6. **SaaS trial-tier abuse causing resource exhaustion.** One instance absorbs disproportionate free-tier traffic and degrades; cycled out while rate-limiting is fixed upstream.
7. **Feature-flag rollout crash-looping a subset of instances.** Flag causes a startup crash on some but not all instances (race condition); crash-looping ones are auto-replaced and paged for investigation.
8. **Multi-tenant DB connection pool starvation.** An instance holds stale/leaked pooled connections; replaced, new instance starts with a clean pool.
9. **SSO integration outage on one node's cached IdP metadata.** Stale metadata causes auth failures from one instance; replacement bootstrap pulls fresh config via SSM.
10. **Log-forwarding agent crash silently degrading an instance.** Sidecar crash causes upstream health probe timeouts; instance replaced, logs preserved via S3 archive first.

Fintech & Banking

1. **Mid-transaction instance failure during ACH batch processing.** Graceful drain (not hard kill) ensures in-flight transactions complete or are cleanly requeued, critical for financial consistency.
2. **Fraud-scoring service instance hangs on a malformed payload.** Health check fails, instance replaced, malformed request logged to CloudTrail for audit.

3 **PCI-DSS compliance requirement for prompt patching.** SSM Automation bootstrap on every replacement ensures new instances always launch with the latest AMI/patch level, satisfying continuous-compliance audits.

4 **Ledger-service instance clock drift causing reconciliation errors.** Health check (or custom NTP-drift check) catches the node before it writes bad timestamps; replaced.

5 **Regulatory audit trail requirement.** CloudTrail + CloudWatch Logs capture every lifecycle event (who/what/when an instance was cycled), satisfying SOC 2 / audit evidence requests.

6 **Trading-hours SLA (sub-3-minute recovery).** RTO < 3 min target ensures a degraded instance during market hours is replaced before it materially affects order execution.

7 **Session-token leakage prevention on terminated instances.** Drain step archives/scrubs session state to S3 rather than letting a terminated instance's local disk (with tokens in memory) just vanish uncleanly.

8 **KYC document-processing instance stuck on a corrupt upload.** Health check timeout catches the hang; instance replaced, stuck job requeued to a healthy node.

9 **Wire-transfer approval service failover.** Multi-AZ + auto-replace ensures no single points of failure in a workflow where downtime has direct financial/legal consequences.

10 **Insider-threat/config-drift detection.** If an instance's config diverges from the Launch Template baseline (manual SSH changes), it can be flagged and cycled back to a known-good state.

Healthcare & Life Sciences

1 **Patient portal instance degrading during appointment booking windows.** Self-healing keeps the portal available during peak morning booking traffic without manual scaling intervention.

2 **HL7/FHIR interface engine instance memory bloat.** Long-running message-processing node leaks memory over a shift; replaced automatically before it drops messages.

3 **HIPAA audit logging requirement.** CloudTrail records every instance lifecycle action, supporting HIPAA technical safeguards documentation.

4 **Telehealth session-handling instance failure mid-visit.** Drain-before-terminate reduces (though doesn't eliminate) risk of abrupt session drops during a live video consult.

5 **Lab-results processing pipeline node stuck on a malformed HL7 message.** Health check catches the hang, instance replaced, poison message isolated for manual review.

6. **After-hours on-call reduction for clinical IT staff.** Routine instance failures at 3am are resolved automatically; SNS notifies but doesn't require action, reducing burnout for a small ops team.
7. **Disaster-recovery drill for a hospital system.** Multi-AZ self-healing fleet is the baseline building block demonstrated in DR tabletop exercises.
8. **Patch compliance for a legacy Windows EC2 fleet running clinical software.** Every replacement re-bootstraps from a patched AMI via SSM, avoiding manual patch-Tuesday fleet-wide reboots.
9. **PHI data residency enforcement.** Instances always relaunch in approved subnets/AZs per the Launch Template, preventing accidental cross-region drift.
10. **Insurance-claims batch job instance crash.** Mid-batch failure triggers drain + S3 archive of partial state, replacement instance resumes from checkpoint.

Media, Streaming & Gaming

1. **Live-stream ingest node overload during a major sports event.** Unhealthy ingest instances are cycled out in real time as viewership spikes.
2. **Game server instance desync causing player disconnects.** Health check (custom TCP/game-protocol probe) detects a desynced server and replaces it, with graceful player-session draining.
3. **Video transcoding worker stuck on a corrupt input file.** Health check timeout catches the hang; job requeued to a fresh instance.
4. **CDN origin server instance serving stale cache.** Cycled automatically if cache-invalidation health checks fail.
5. **Esports tournament infrastructure reliability.** Self-healing fleet is the resilience layer behind a match server pool where downtime during a broadcast is highly visible and reputationally costly.
6. **Ad-insertion service instance failure mid-stream.** Replaced without interrupting the broader stream, since only the failed instance's segment is affected.
7. **User-generated content upload service instance disk-full condition.** Health check (disk-space-aware) catches it before uploads start failing, instance replaced with clean storage.
8. **Matchmaking service instance queue backup.** A hung matchmaking node is detected via custom health metric and cycled, preventing player queue times from silently growing.
9. **Post-launch day-one patch traffic surge.** New game release causes massive concurrent load; self-healing keeps unhealthy nodes from compounding an already-stressed fleet.

10 **Regional latency degradation on one instance.** CloudWatch alarm on p99 latency triggers investigation/replacement even before hard health-check failure.

Travel & Hospitality

1 **Airline booking engine instance hang during a fare-sale event.** High-concurrency booking traffic degrades one node; replaced before it causes double-bookings or timeouts.

2 **Hotel PMS integration instance losing connectivity to a property system.** Health check catches the broken integration, instance cycled, connection re-established on the fresh node.

3 **Flight-status API instance serving stale data during an IRROPS event.** Irregular operations (e.g., mass cancellations) — self-healing ensures the API tier stays responsive when it matters most.

4 **Loyalty-points calculation service instance crash mid-batch.** Drain + S3 archive preserves partial calculation state for reprocessing.

5 **Check-in kiosk backend instance overload during a holiday travel peak.** Unhealthy backend nodes replaced automatically ahead of a known traffic pattern (paired with scheduled scaling).

6 **Dynamic pricing engine instance stuck in a bad state.** Serving incorrect prices; custom health check catches logical (not just network) failures.

7 **Multi-region DR for a booking platform.** Self-healing fleet in each region is the unit of resilience that a broader multi-region failover strategy builds on top of.

8 **Third-party GDS connector instance timeout cascade.** One instance's hung GDS (global distribution system) connections are isolated and replaced before they exhaust a shared connection pool.

Logistics & Supply Chain

1 **Warehouse management system instance losing sync with barcode scanners.** Health check (custom) detects desync, instance replaced during a shift change window.

2 **Route-optimization batch job instance running out of memory on a large job.** Replaced, job requeued to a fresh instance with clean memory.

3 **Real-time package-tracking API instance degradation during peak shipping season.** E.g., holiday e-commerce surge — self-healing keeps tracking updates flowing.

4. **EDI processing node stuck on a malformed file.** Health check timeout isolates the hang, instance replaced, bad file quarantined.
5. **Fleet-telemetry ingestion instance overload from IoT sensor burst.** Unhealthy ingestion nodes cycled as sensor data volume spikes.
6. **Cold-chain monitoring alert service instance failure.** Critical for pharma/food logistics; self-healing minimizes the window where temperature-breach alerts could be missed.
7. **Customs/compliance document processing instance crash.** Drain preserves in-flight document state before replacement.

Government & Public Sector

1. **Citizen services portal instance failure during tax season or benefits enrollment period.** Self-healing absorbs predictable seasonal load spikes without manual intervention.
2. **FedRAMP/continuous-monitoring patch compliance.** Every instance replacement launches from an approved, patched AMI, supporting continuous ATO (Authority to Operate) requirements.
3. **Emergency alert system backend instance degradation.** High-availability requirement (life-safety adjacent) makes automatic sub-3-minute recovery directly relevant.
4. **Public records request processing instance stuck on a large query.** Health check catches the hang, instance replaced, query killed and logged.
5. **Election-night results reporting infrastructure.** A highly visible, short-duration, high-traffic event where self-healing under load is the difference between "site is fine" and "site is down" headlines.
6. **Unemployment benefits portal surge during an economic downturn.** Self-healing fleet plus ASG scaling handles unpredictable demand spikes without a war-room scaling event.

AdTech, Marketing & Data

1. **Real-time bidding (RTB) instance latency degradation.** Ad exchanges have strict SLA latency requirements (~100ms); an instance breaching that is caught by a latency-based CloudWatch alarm and cycled before it hurts the exchange relationship.
2. **Marketing email-send worker stuck mid-batch.** Health check timeout catches the hang, batch resumes on a fresh instance, avoiding duplicate sends.

3 **Clickstream ingestion instance dropping events under load.** Unhealthy ingestion nodes replaced, minimizing data loss for downstream analytics.

4 **A/B testing assignment service instance serving inconsistent variants.** Custom health check on assignment-consistency catches a logic-level failure, not just a network one.

5 **Attribution pipeline instance crash during a campaign launch.** Drain preserves partial attribution data before replacement.

6 **Customer data platform (CDP) instance connection pool exhaustion during a major campaign send.** Replaced automatically instead of manually paged at 2am.

Internal DevOps & Enterprise IT

1 **CI/CD runner instance stuck on a hung build.** Health check timeout recycles the runner, unblocking the pipeline without manual intervention.

2 **Internal wiki/Confluence-alternative instance degradation.** Lower-stakes but still benefits from the same pattern, demonstrating the architecture generalizes beyond customer-facing systems.

3 **VPN/bastion host instance failure.** Self-healing keeps internal remote-access infrastructure available (with tighter security-group scoping than public-facing tiers).

4 **Internal API gateway instance memory leak from a chatty microservice.** Cycled automatically before it degrades other internal consumers.

5 **Monitoring/observability stack self-hosting.** E.g., a self-hosted Prometheus/Grafana instance — ironic but real: the monitoring system itself needs to be resilient, and this pattern applies recursively.

6 **On-call rotation reduction as an explicit engineering goal.** The project's core pitch to an eng org: fewer 2am pages for "just restart the instance" class incidents.

7 **Compliance-driven immutable infrastructure adoption.** Replacing instances via SSM-bootstrapped Launch Templates (rather than patching in place) supports an org-wide push toward immutable infra.

8 **Cost optimization via right-sized min/max ASG bounds combined with self-healing.** Avoids over-provisioning "just in case" since unhealthy capacity is replaced rather than needing standby buffers.

9 **Chaos-engineering validation target.** The fleet itself becomes the system under test in chaos experiments (e.g., randomly terminating instances to verify the self-healing loop actually works, à la Chaos Monkey).

10 **Blue/green or canary deployment safety net.** If a canary batch of instances fails health checks post-deploy, they're cycled automatically while the deploy pipeline gets a clear signal to halt.

11 **Internal ML feature-store instance degradation.** Feature-serving nodes for ML pipelines benefit from the same recovery loop as customer-facing services.

12 **Legacy monolith lift-and-shift to EC2 with limited refactoring budget.** Self-healing gives a legacy app HA characteristics without rewriting it for containers/serverless.

13 **Log-shipping agent failure silently causing observability gaps.** Instance-level health check tied to agent liveness catches "app is fine but we're blind" scenarios.

Cross-Cutting & Architectural Scenarios

1 **Instance-level security patch (CVE) rollout.** Updating the Launch Template's AMI and letting natural instance cycling (or a forced refresh) roll the patch out fleet-wide gradually.

2 **Spot Instance interruption handling.** The same lifecycle-hook pattern (adapted for EC2_SPOT_INSTANCE_TERMINATION) can gracefully drain Spot capacity before AWS reclaims it, not just handle health-check failures.

3 **Cross-region disaster recovery drill.** Demonstrating that a self-healing fleet in a secondary region can absorb failover traffic from a primary-region outage.

4 **Zero-downtime AMI rotation for base-image security hardening.** New hardened AMI rolled into the Launch Template; instances cycle onto it over time without a maintenance window.

5 **Capacity-aware self-healing during a dependency outage.** E.g., a downstream DB is slow — distinguishing "my instance is broken" from "a dependency is broken" so the fleet doesn't churn-replace every instance chasing a problem replacement can't fix (an important design consideration, not just a use case).

6 **Multi-tenant SaaS "noisy neighbor" isolation at the infra layer.** Reinforces #11 at an architectural level: unhealthy-node replacement as a blast-radius containment tool, not just an uptime tool.

7 **Interview/portfolio narrative.** "I designed and (partially) built a production-grade self-healing compute tier using ASG lifecycle hooks, EventBridge, and Lambda, targeting sub-3-minute RTO and 99.95% availability" is a strong, specific talking point backed by an actual architecture doc.

8 **Cost-of-downtime justification for stakeholders.** RTO/availability targets in this project's README give a concrete number ("99.95% = ~4.4 hrs/year downtime budget") to anchor a business case for investing in automation.

9 **Baseline for a service-mesh or Kubernetes migration comparison.** This EC2-native pattern is a useful "before" state to contrast against if the org later evaluates EKS/ECS with built-in self-healing, showing you understand the tradeoffs of each.

10 **Template for other stateless-tier self-healing projects.** The same lifecycle-hook + Lambda + SSM pattern generalizes to any stateless EC2 fleet (internal tools, batch workers, API tiers) beyond this specific capstone, making this a reusable reference architecture.

> *Note: many of these (custom health checks on latency/logic/queue-depth rather than just TCP/HTTP reachability) require extending the health-check layer beyond the base ALB target-group check described in ARCHITECTURE.md. Worth flagging in PLAN.md if any of these get prioritized.*